

Count Data Models

From Poisson to Zero-Inflated Models

StatHarbor Analytics

Why Count Models Matter

- ▶ Many real-world outcomes are counts:
 - ▶ Number of hospital visits
 - ▶ Number of adverse events
 - ▶ Number of insurance claims
- ▶ These are:
 - ▶ Non-negative integers
 - ▶ Often skewed
 - ▶ Often heavy in zeros
- ▶ Linear regression is inappropriate because it assumes normally distributed errors and constant variance.

Core Modeling Challenge

Fundamental problem

How do we model:

- ▶ Mean structure
- ▶ Variability
- ▶ Excess zeros

Key idea

Different models encode different assumptions about the data generating process.

Poisson Distribution

- ▶ $Y \sim \text{Poisson}(\lambda)$ with probability mass function $\Pr(Y = k) = \frac{\lambda^k e^{-\lambda}}{k!}$.
- ▶ Mean equals variance:

$$E[Y] = \text{Var}(Y) = \lambda.$$

Interpretation

A single rate parameter controls both the mean and the variability. This equi-dispersion implies that Poisson regression can underestimate variability when data are overdispersed.

Poisson Regression Model

$$\log(\lambda_i) = X_i\beta$$

- ▶ The log link ensures positive expected counts.
- ▶ Covariates act multiplicatively on the rate: a one unit change in x_j multiplies λ by e^{β_j} .

Interpretation

Exponentiated coefficients e^{β_j} represent rate ratios relative to the baseline.

When Poisson Works

- ▶ The conditional mean roughly equals the conditional variance.
- ▶ Data do not exhibit an excess of zeros beyond what the Poisson distribution would predict.
- ▶ Events occur independently within a fixed exposure period.

Limitations of Poisson

Overdispersion

When $\text{Var}(Y) > E[Y]$, the equi-dispersion assumption is violated and the model underestimates standard errors, leading to inflated Type I error rates.

- ▶ Real data often show variability larger than predicted by the Poisson model.
- ▶ Overdispersion can be due to unobserved heterogeneity, omitted covariates, or clustering.

Negative Binomial Motivation

- ▶ Many count datasets are overdispersed relative to Poisson.
- ▶ The negative binomial (NB) model introduces an extra dispersion parameter to accommodate this overdispersion.
- ▶ The NB distribution can be derived as a Poisson-Gamma mixture; this interpretation treats the Poisson rate λ as a Gamma-distributed latent variable capturing unobserved heterogeneity.

Key concept

Treating λ as random induces additional variance in Y beyond the Poisson assumption, naturally producing overdispersion.

Negative Binomial Variance Functions

Different parameterizations exist for the variance of the NB distribution:

- ▶ **Inverse relationship:** Many authors (including McCullagh and Nelder) define the variance as

$$\text{Var}(Y) = \mu + \frac{\mu^2}{\alpha},$$

which explicitly reflects the Poisson-Gamma mixture.

- ▶ **Direct relationship:** Nelder's preferred parameterization relates the dispersion parameter directly to the square of the mean:

$$\text{Var}(Y) = \mu + \alpha\mu^2.$$

In both cases, α (or $1/\alpha$) quantifies the amount of overdispersion—larger values imply greater departure from the Poisson.

Negative Binomial Model and GLM Status

- ▶ The NB model can be expressed within the generalized linear model (GLM) framework if the dispersion parameter α is treated as known (ancillary).
- ▶ When α is estimated from the data, the estimation uses full maximum likelihood; technically the model is no longer a one-parameter exponential family GLM.
- ▶ As $\alpha \rightarrow 0$, the NB model approaches the Poisson model, so the Poisson is a special case of the NB model.

Variance Structure

The NB regression uses the same log link as the Poisson but models

$\text{Var}(Y) = \mu + \alpha\mu^2$ or $\text{Var}(Y) = \mu + \frac{\mu^2}{\alpha}$, depending on the parameterization.

When to Use the Negative Binomial

- ▶ Evidence of overdispersion: the variance substantially exceeds the mean after controlling for covariates.
- ▶ No compelling structural reason for excess zeros—i.e., zeros arise from the same stochastic process as positive counts.
- ▶ Model comparisons (e.g., likelihood ratio tests) favor NB over Poisson.

Why Zero Inflation?

- ▶ Many datasets have more zeros than expected under standard Poisson or NB models (e.g., structural zeros such as individuals who never participate in the process).
- ▶ In such cases, the excess zeros may be generated by a separate process and should be modeled independently.
- ▶ Zero-inflated models accommodate two data generating mechanisms: one that produces only zeros and another that produces counts.

Conceptual Framework

Mixture model

Let p denote the probability of an observation belonging to the always-zero class. With probability $(1 - p)$ the observation arises from a count model with mean μ . The probability mass function is

$$\Pr(Y = 0) = p + (1 - p)f_Y(0|\mu), \quad \Pr(Y = k > 0) = (1 - p)f_Y(k|\mu),$$

where $f_Y(\cdot)$ is a Poisson or Negative Binomial mass function. This model has two parts: a logistic regression for the zero versus non-zero probability, and a count model for the positive counts.

Zero-Inflated Poisson (ZIP) Model

- ▶ **Structure:** Combination of a Bernoulli (logistic) model and a Poisson count model.
- ▶ **Probabilities:**

$$\Pr(Y = 0) = p + (1 - p)e^{-\lambda}, \quad \Pr(Y = k > 0) = (1 - p)\frac{\lambda^k e^{-\lambda}}{k!}.$$

- ▶ **When to use:**
 - ▶ Data show many zeros that cannot be explained by covariates alone.
 - ▶ Dispersion arises primarily from structural zeros rather than heterogeneity.
- ▶ ZIP models fit two sets of coefficients: one for the logit (zero inflation) and one for the log-Poisson (count) components.

Zero-Inflated Negative Binomial (ZINB) Model

- ▶ **Structure:** Like ZIP but uses a Negative Binomial distribution in the count component. The probability of zero is

$$\Pr(Y = 0) = p + (1 - p)f_{\text{NB}}(0|\mu, \theta),$$

where f_{NB} is the NB pmf with dispersion parameter θ .

- ▶ **Interpretation:** Allows both structural zeros and overdispersion in the positive counts. Includes two sets of covariates: one for the zero-inflation process and one for the count process.
- ▶ **GLM status:** ZINB is not a member of the exponential family; it is a finite mixture model and hence lies outside the standard GLM framework.
- ▶ **When to use:** When there is evidence of both excess zeros and overdispersion beyond what the Poisson can handle.

Model Selection Strategy

- ▶ Step 1: Examine dispersion. Compare the mean and variance or use overdispersion diagnostics to decide between Poisson and NB.
- ▶ Step 2: Investigate the prevalence of zeros. If zeros are far more common than predicted by Poisson or NB, consider zero-inflated models.
- ▶ Step 3: Fit candidate models and compare via information criteria (AIC/BIC), likelihood ratio tests or Vuong tests.

Diagnostic Tools

- ▶ Mean-variance plots: plot observed variance versus mean across groups.
- ▶ Residual diagnostics: compare deviance or Pearson residuals against fitted values.
- ▶ Information criteria: AIC or BIC for nested and non-nested models.
- ▶ Vuong test: compares non-nested models such as ZIP and Poisson.

Summary Table

Model	Overdispersion	Structural zeros	GLM?
Poisson	No	No	Yes
Negative Binomial	Yes	No	Yes/No*
ZIP	No	Yes	No
ZINB	Yes	Yes	No

*NB is a GLM only if the dispersion parameter is treated as known; otherwise estimation uses full likelihood.

Real-World Applications

- ▶ **Healthcare utilization:** modeling the number of doctor visits or hospital admissions.
- ▶ **Insurance claims:** predicting the number of claims filed by policy holders.
- ▶ **Epidemiology:** counts of disease incidence, adverse events or hospitalizations.

Practical Insight

Key message

Choosing an inappropriate model leads to biased inference: underestimated uncertainty, incorrect rate ratios and misleading conclusions. Always diagnose dispersion and zero inflation before settling on a model.

References

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